

Homework 8

Due Friday, April 25

The action of the lowering operators corresponding to the simple roots on the weights of the 3 of SU(3) is

$$E_{-\alpha^1}|_1\rangle = \frac{1}{\sqrt{2}}|_3\rangle, \quad E_{-\alpha^2}|_3\rangle = \frac{1}{\sqrt{2}}|_2\rangle$$

(all other applications being zero).

1. (10 pts.) Recall that the highest-weight state of the 10 of SU(3) is

$$|HW\rangle = |_1\rangle|_1\rangle|_1\rangle.$$

(a) Find the states

$$\begin{aligned} |A\rangle &= \mathcal{N}_A E_{-\alpha^1} |HW\rangle, \\ |B\rangle &= \mathcal{N}_B E_{-\alpha^1} |A\rangle, \end{aligned}$$

including positive normalization constants $\mathcal{N}_A$ and $\mathcal{N}_B$ chosen so that the norms of $|A\rangle$ and $|B\rangle$ are unity.

(b) Recall that any state $|D\rangle$ in 10 can be written as $|D\rangle = D^{ijk}|_i\rangle|_j\rangle|_k\rangle$ with a completely symmetric tensor D^{ijk} . Find these tensors for the states $|HW\rangle$, $|A\rangle$, and $|B\rangle$.

(c) Let

$$h^i_j = \text{diag}(1, 1, -2)$$

be a tensor corresponding to an element in 8. Find the singlet S (there is only one) that can be made from D^{ijk} , its conjugate $\bar{D}_{lmn}$, and h^p_q (using one copy of each) and compute the value of S for the three states listed in part (b).

2. (10 pts.) Decompose the following tensor products into direct sums of irreps of SU(3) using Young tableaux:

(a) $(2, 1) \otimes \bar{3}$, where $(2, 1)$ is one of the 15-dimensional irreps [the (n, m) notation for the irreps is defined in eq. (9.27) of the textbook],

(b) $\bar{6} \otimes 6$.

The dimensions of the irreps are given by eq. (10.43) of the textbook.